

Prioritizing our efforts

We’re making progress toward our goal of sourcing only recycled and renewable materials for our products: In 2024, 24 percent of the materials contained in products shipped came from recycled or renewable sources.²⁸

We’ve focused our efforts on 15 priority materials based on a broad range of environmental, social, and supply chain impacts. Each are outlined in detail in our [Material Impact Profiles](#) white paper.³⁰ Our priority materials consist of aluminum, cobalt, copper, glass, gold, lithium, paper, plastics, rare earth elements, steel, tantalum, tin, titanium, tungsten, and zinc, and they account for 87 percent of the total product mass shipped to our customers in 2024.

Maintaining our standards for recycled and renewable materials is essential to our journey to create a circular supply chain. Our requirements are based on international standards for recycled content and responsible resource management. By requiring certification to these standards, we’re able to confirm that a material has been recycled or comes from a renewable source — one that can continually produce without depleting the earth’s natural resources. We approach materials from new sources with the same rigor, evaluating each one for the safety of the material’s chemistry. This process allows us to scale our use of materials that are better for the environment and safer for use in our products. Recycled material is certified by third parties to a recycled content standard that conforms with ISO 14021. Total recycled content numbers also include supplier-reported recycled content checked by Apple but not third-party certified.

Our teams are overcoming obstacles to creating closed loop supply chains, including material performance and traceability. This is possible through our work with a diverse group of partners stretching from suppliers to metallurgists to product designers. For example, we were able to design an alloy that meets our rigorous design performance standards and contains 100 percent recycled aluminum.

Barriers to our progress remain — including challenges within our control and those outside our direct influence. Addressing these requires a collective response. Through collaboration across the value chain, we can achieve impact felt beyond our business. The supply chains we’re helping innovate serve more than just our product needs — they help promote the availability of competitively priced, quality recycled and renewable materials across geographies.

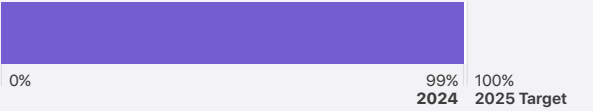
Goals

Transition the following to 100 percent recycled by the end of 2025:²⁹

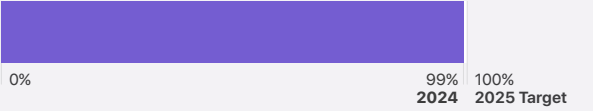
- Cobalt in all Apple-designed batteries
- Tin soldering in all Apple-designed rigid and flexible printed circuit boards
- Gold plating in all Apple-designed rigid and flexible printed circuit boards
- Rare earth elements in all magnets across all products

Progress

Recycled gold



Recycled tin



Recycled cobalt



Recycled rare earth elements



Key challenges to developing circular supply chains



Technical properties

While many recycled or renewable materials have indistinguishable technical properties, some may differ from the conventional material. This needs to be accounted for during product design and manufacturing. For example, select recycled plastics differ in properties from other plastics. The composition of other recycled materials can also be impacted by some level of contamination during the recycling process.



Availability and access

The supply of recycled and renewable materials can be constrained by the limited availability of recoverable material or production of renewable content. When supply exists in some locations around the world, new suppliers need to be incorporated into supply chains for the material to be accessed. Technological limitations in recovering materials from complex waste streams can pose challenges to efficiency and effectiveness.



Supply chains

Recycled or renewable content may not be easily accessible on the market, requiring the development of new supply chains.



Traceability

Information about the source of materials — whether mined, recycled, or renewable — might not be readily available.



Scale

Materials for a single component can come from hundreds of different suppliers, requiring exponentially more effort as we scale the use of high-quality recycled or renewable materials across components and products.



Regulatory barriers

Transboundary movement regulations — created to establish critically important community and environmental protections — can have the unintended consequence of inhibiting material recovery and movement to recyclers or refiners for use in new products. Apple supports the U.S. ratification of the Basel Convention to better support recycling and recovery with this key geography. To learn about our support of other policies that enable circular supply chains while improving social and environmental protections, read the [Apple 2030 policy platform](#).